GAYATRI VIDYA PARISHAD COLLEGE FOR DEGREE AND PG COURSES (A)

Department of Engineering & Technology

Scheme of Valuation

Course: Engineering Mathematics-II
Semester: II
Examination: MID-II
Maximum Marks: 30

PART – A (3 × 2 = 6 Marks)

Q.1(a) Find the orthogonal trajectories of the given family of parabolas. *(2 Marks)*

• Formation of differential equation by eliminating the parameter – **1 Mark**
• Replacement of slope by negative reciprocal and final orthogonal trajectory – **1 Mark**

Q.1(b) Solve the given differential equation. *(2 Marks)*

• Identification of the appropriate method – **1 Mark**
• Correct solution – **1 Mark**

Q.1(c) Find the Laplace transform of the given function. *(2 Marks)*

• Application of Laplace transform formula/properties – **1 Mark**
• Correct transformed result – **1 Mark**

PART – B (2 × 12 = 24 Marks)

Q.2(a) Solve the given differential equation. *(6 Marks)*

• Identification of the method – **1 Mark**
• Correct formulation – **2 Marks**
• Integration/solution steps – **2 Marks**
• Final answer – **1 Mark**

Q.2(b) Solve the given differential equation. *(6 Marks)*

• Appropriate method – **1 Mark**
• Intermediate steps – **3 Marks**
• Simplification and final solution – **2 Marks**

OR

Q.3(a) Construct the differential equation for the orthogonal trajectories of the given family of curves. *(6 Marks)*

- Elimination of the parameter – **2 Marks**
- Formation of differential equation – **1 Mark**
- Replacement of slope by negative reciprocal – **1 Mark**
- Integration and final orthogonal trajectory – **2 Marks**

Q.3(b) A body is originally at 80°C and cools to 60°C in 20 minutes. Find the temperature after 40 minutes. *(6 Marks)*

- Formulation using Newton's Law of Cooling – **2 Marks**
- Determination of the constant – **2 Marks**
- Calculation of temperature after 40 minutes – **2 Marks**

Q.4(a) Solve the given differential equation. *(6 Marks)*

- Identification of method – **1 Mark**
- Solution procedure – **3 Marks**
- Correct final answer – **2 Marks**

Q.4(b) Solve by the Method of Variation of Parameters. *(6 Marks)*

- Complementary function – **2 Marks**
- Particular integral using variation of parameters – **3 Marks**
- General solution – **1 Mark**

OR

Q.5(a) Using the Convolution Theorem, find the required inverse Laplace transform. *(6 Marks)*

- Laplace transforms of component functions – **2 Marks**
- Application of Convolution Theorem – **2 Marks**
- Evaluation of convolution integral – **1 Mark**
- Final answer – **1 Mark**

Q.5(b) Solve the given differential equation using Laplace Transform. *(6 Marks)*

- Laplace transformation of the equation – **2 Marks**
- Solution in the transform domain – **2 Marks**
- Inverse Laplace transform – **1 Mark**
- Final solution satisfying the given conditions – **1 Mark**

General Instructions to Examiners

1. Award marks step-wise for every correct mathematical step.
2. Give due credit for correct methodology even if minor arithmetic errors occur.
3. Accept any standard and mathematically valid alternative method.
4. Deduct marks only for conceptual or computational errors.
5. Ensure that the total marks awarded do not exceed the prescribed maximum.